Diagrammi UML e Architettura dei Package

1. Architettura dei Package e Dipendenze («import»)

Diagram Error: No diagram type detected matching given configuration for text: flowchart TD subgraph ClientModule ["Modulo Client (Maven)"] PkgClient["Client"] PkgUtility["utility"] PkgClientDto["dto"] end subgraph ServerModule ["Modulo Server (Maven)"] PkgServer["server"] PkgTree["tree"] PkgData["data"] PkgDatabase["database"] PkgServerDto["dto"] end PkgClient -.->|"«import»"| PkgUtility PkgClient -.->|"«import»"| PkgClientDto PkgServer -.->|"«import»"| PkgTree PkgServer -.->|"«import»"| PkgData PkgServer -.->|"«import»"| PkgDatabase PkgServer -.->|"«import»"| PkgServerDto PkgTree -.->|"«import»"| PkgData PkgTree -.->|"«import»"| PkgServerDto PkgTree -.->|"«import»"| PkgServer PkgData -.->|"«import»"| PkgDatabase

2. Diagramma UML delle Classi — Package `data`

Diagram Error: No diagram type detected matching given configuration for text: classDiagram class Attribute { <> -long serialVersionUID$ -String name -int index #Attribute(name: String, index: int) +getName() String +getIndex() int +toString() String } class ContinuousAttribute { -long serialVersionUID$ +ContinuousAttribute(name: String, index: int) } class DiscreteAttribute { -long serialVersionUID$ -Set~String~ values +DiscreteAttribute(name: String, index: int, values: String[]) +getNumberOfDistinctValues() int +iterator() Iterator~String~ } class Data { -List~Example~ data -int numberOfExamples -List~Attribute~ explanatorySet -ContinuousAttribute classAttribute +Data(tableName: String) -isNumericType(sqlType: int) boolean +getNumberOfExamples() int +getNumberOfExplanatoryAttributes() int +getClassValue(exampleIndex: int) Double +getExplanatoryValue(exampleIndex: int, attributeIndex: int) Object +getExplanatoryAttribute(index: int) Attribute +getClassAttribute() ContinuousAttribute +toString() String +sort(attribute: Attribute, beginExampleIndex: int, endExampleIndex: int) void -swap(i: int, j: int) void -partition(attribute: DiscreteAttribute, inf: int, sup: int) int -partition(attribute: ContinuousAttribute, inf: int, sup: int) int -quicksort(attribute: Attribute, inf: int, sup: int) void +main(args: String[])$ void } class TrainingDataException { -long serialVersionUID$ +TrainingDataException(message: String) +TrainingDataException(message: String, cause: Throwable) } Attribute <|-- ContinuousAttribute Attribute <|-- DiscreteAttribute Iterable <|.. DiscreteAttribute Serializable <|.. Attribute Exception <|-- TrainingDataException Data *-- "0..*" Example : data Data *-- "1..*" Attribute : explanatorySet Data *-- "1" ContinuousAttribute : classAttribute

3. Diagramma UML delle Classi — Package `database`

Diagram Error: No diagram type detected matching given configuration for text: classDiagram class DbAccess { -String DRIVER_CLASS_NAME$ -String DBMS$ -int PORT$ -String SERVER -String DATABASE -String USER_ID -String PASSWORD -Connection conn +DbAccess() +initConnection() void +getConn() Connection +getConnection() Connection +closeConnection() void } class Column { -String name -String type ~Column(name: String, type: String) +getColumnName() String +isNumber() boolean +toString() String } class Example { -List~Object~ example +Example() +add(o: Object) void +get(i: int) Object +compareTo(ex: Example) int +toString() String +iterator() Iterator~Object~ } class TableSchema { -List~Column~ tableSchema +TableSchema(db: DbAccess, tableName: String) +getNumberOfAttributes() int +getColumn(index: int) Column +iterator() Iterator~Column~ } class TableData { -DbAccess db +TableData(db: DbAccess) +getTransazioni(table: String) List~Example~ +getDistinctColumnValues(table: String, column: Column) Set~Object~ } class DatabaseConnectionException { -long serialVersionUID$ +DatabaseConnectionException(message: String) +DatabaseConnectionException(message: String, cause: Throwable) } class EmptySetException { -long serialVersionUID$ +EmptySetException(message: String) +EmptySetException(message: String, cause: Throwable) } Iterable <|.. Example Comparable <|.. Example Iterable <|.. TableSchema Exception <|-- DatabaseConnectionException Exception <|-- EmptySetException TableSchema *-- "1..*" Column : tableSchema TableData o-- "1" DbAccess : db

4. Diagramma UML delle Classi — Package `tree`

Diagram Error: No diagram type detected matching given configuration for text: classDiagram class Node { <> -long serialVersionUID$ -int idNodeCount$ -int idNode -int beginExampleIndex -int endExampleIndex -double variance ~Node(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int) ~getIdNode() int ~getBeginExampleIndex() int ~getEndExampleIndex() int ~getVariance() double ~getNumberOfChildren()* int +toString() String } class LeafNode { -long serialVersionUID$ -Double predictedClassValue ~LeafNode(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int) ~getPredictedClassValue() Double ~getNumberOfChildren() int +toString() String } class SplitNode { <> -long serialVersionUID$ -Attribute attribute #List~SplitInfo~ mapSplit -double splitVariance ~SplitNode(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int, attribute: Attribute) #setSplitInfo(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int, attribute: Attribute)* void ~testCondition(value: Object)* int ~getAttribute() Attribute ~getVariance() double ~getNumberOfChildren() int ~getSplitInfo(child: int) SplitInfo ~formulateQuery() String +compareTo(o: SplitNode) int +toString() String } class SplitInfo { -long serialVersionUID$ -Object splitValue -int beginIndex -int endIndex -int numberChild -String comparator ~SplitInfo(splitValue: Object, beginIndex: int, endIndex: int, numberChild: int) ~SplitInfo(splitValue: Object, beginIndex: int, endIndex: int, numberChild: int, comparator: String) ~getBeginIndex() int ~getEndIndex() int ~getSplitValue() Object ~getNumberChild() int ~getComparator() String +toString() String } class DiscreteNode { -long serialVersionUID$ ~DiscreteNode(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int, attribute: DiscreteAttribute) #setSplitInfo(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int, attribute: Attribute) void ~testCondition(value: Object) int +toString() String } class ContinuousNode { -long serialVersionUID$ ~ContinuousNode(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int, attribute: ContinuousAttribute) #setSplitInfo(trainingSet: Data, beginExampleIndex: int, endExampleIndex: int, attribute: Attribute) void ~testCondition(value: Object) int +toString() String } class RegressionTree { -long serialVersionUID$ -Node root -RegressionTree[] childTree -RegressionTree() +RegressionTree(trainingSet: Data) -learnTree(trainingSet: Data, begin: int, end: int, numberOfExamplesPerLeaf: int) void -isLeaf(trainingSet: Data, begin: int, end: int, numberOfExamplesPerLeaf: int) boolean -determineBestSplitNode(trainingSet: Data, begin: int, end: int) SplitNode +printRules() void -printRules(current: String) void +getRules() String +printTree() void +toString() String +predictClass(out: ObjectOutputStream, in: ObjectInputStream) Double +navigate(path: List~Integer~) PredictionStepDto +salva(nomeFile: String) void +carica(nomeFile: String)$ RegressionTree } Node <|-- LeafNode Node <|-- SplitNode SplitNode <|-- DiscreteNode SplitNode <|-- ContinuousNode Comparable <|.. SplitNode Serializable <|.. Node Serializable <|.. SplitInfo Serializable <|.. RegressionTree RegressionTree *-- "1" Node : root RegressionTree *-- "0..*" RegressionTree : childTree SplitNode *-- "1..*" SplitInfo : mapSplit SplitNode o-- "1" Attribute : attribute

5. Diagramma UML delle Classi — Package `server`

Diagram Error: No diagram type detected matching given configuration for text: classDiagram class ServerApplication { +ServerApplication() +main(args: String[])$ void } class TreeService { -Map~String,RegressionTree~ activeTrees +TreeService() +learnFromDb(tableName: String) TreeResponseDto +loadFromFile(tableName: String) TreeResponseDto +predictStep(tableName: String, path: List~Integer~) PredictionStepDto +clearCache() void } class TreeController { -TreeService treeService +TreeController(treeService: TreeService) +learn(request: TreeRequestDto) ResponseEntity~TreeResponseDto~ +load(request: TreeRequestDto) ResponseEntity~TreeResponseDto~ +predict(request: PredictRequestDto) ResponseEntity~PredictionStepDto~ } class GlobalExceptionHandler { +GlobalExceptionHandler() +handleFileNotFound(ex: FileNotFoundException) ResponseEntity +handleDomainExceptions(ex: Exception) ResponseEntity +handleIoAndFormatExceptions(ex: Exception) ResponseEntity +handleGenericException(ex: Exception) ResponseEntity } class MultiServer { -int port +MultiServer(port: int) -run() void } class ServerOneClient { -Socket socket -ObjectInputStream in -ObjectOutputStream out +ServerOneClient(s: Socket) +run() void } class UnknownValueException { -long serialVersionUID$ +UnknownValueException(message: String) +UnknownValueException(message: String, cause: Throwable) } Thread <|-- ServerOneClient Exception <|-- UnknownValueException TreeController --> TreeService : delega business logic TreeService o-- "0..*" RegressionTree : cache activeTrees MultiServer ..> ServerOneClient : crea thread ServerOneClient ..> RegressionTree : esegue

6. Diagramma UML delle Classi — Package `Client` e `utility`

Diagram Error: No diagram type detected matching given configuration for text: classDiagram class ClientApplication { +ClientApplication() +main(args: String[])$ void } class Main { +Main() +main(args: String[])$ void } class TreeRestClientService { -ObjectMapper OBJECT_MAPPER$ -RestClient restClient +TreeRestClientService() +TreeRestClientService(restClientBuilder: Builder, baseUrl: String) +TreeRestClientService(baseUrl: String) +learnTree(tableName: String) TreeResponseDto +loadTree(tableName: String) TreeResponseDto +predictStep(tableName: String, path: List~Integer~) PredictionStepDto } class MainView { -TreeRestClientService treeRestClientService -String currentTableName -List~Integer~ currentPath -TextField tableNameField -Button learnBtn -Button loadBtn -TextArea rulesArea -VerticalLayout stepContainer +MainView(treeRestClientService: TreeRestClientService) -handleLearn() void -handleLoad() void -renderInitialStepState() void -doPredictStep() void -renderStep(step: PredictionStepDto) void -showSuccess(msg: String) void -showError(msg: String) void } class Keyboard { -boolean printErrors$ -int errorCount$ -String current_token$ -StringTokenizer reader$ -BufferedReader in$ -Keyboard() +getErrorCount()$ int +resetErrorCount(count: int)$ void +getPrintErrors()$ boolean +setPrintErrors(flag: boolean)$ void -error(str: String)$ void -getNextToken()$ String -getNextToken(skip: boolean)$ String -getNextInputToken(skip: boolean)$ String +endOfLine()$ boolean +readString()$ String +readWord()$ String +readBoolean()$ boolean +readChar()$ char +readInt()$ int +readLong()$ long +readFloat()$ float +readDouble()$ double } AppShellConfigurator <|.. ClientApplication VerticalLayout <|-- MainView MainView --> TreeRestClientService : chiama REST Main ..> Keyboard : I/O console

7. Diagramma UML delle Classi — Package `dto` (Contratti JSON)

Diagram Error: No diagram type detected matching given configuration for text: classDiagram class TreeRequestDto { <> -long serialVersionUID$ -String tableName +tableName() String } class PredictionStepDto { <> -long serialVersionUID$ -boolean isLeaf -Double predictedValue -String attributeName -List~String~ branchOptions +isLeaf() boolean +predictedValue() Double +attributeName() String +branchOptions() List~String~ } class TreeResponseDto { <> -long serialVersionUID$ -String tableName -String treeRules -PredictionStepDto rootStep +tableName() String +treeRules() String +rootStep() PredictionStepDto } class PredictRequestDto { <> -long serialVersionUID$ -String tableName -List~Integer~ path +tableName() String +path() List~Integer~ } Serializable <|.. TreeRequestDto Serializable <|.. PredictionStepDto Serializable <|.. TreeResponseDto Serializable <|.. PredictRequestDto TreeResponseDto o-- "1" PredictionStepDto : rootStep

8. Diagramma di Sequenza 1: Apprendimento e Navigazione REST Stateless (Spring Boot 8080)

Diagram Error: No diagram type detected matching given configuration for text: sequenceDiagram autonumber actor Client as Client HTTP / cURL participant Controller as TreeController (8080) participant Service as TreeService participant DB as MySQL MapDB participant Tree as RegressionTree participant Disk as File System (.dmp) Client->>Controller: POST /api/trees/learn {"tableName": "provaC"} Controller->>Service: learnFromDb("provaC") Service->>DB: Query dati tabella 'provaC' DB-->>Service: Recordset tuple di training Service->>Tree: new RegressionTree(data) Service->>Disk: tree.salva("provaC.dmp") Service->>Service: activeTrees.put("provaC", tree) Service->>Tree: tree.navigate([]) Tree-->>Service: PredictionStepDto (radice: attributeName, branchOptions) Service-->>Controller: TreeResponseDto ("provaC", treeRules, rootStep) Controller-->>Client: 200 OK (TreeResponseDto JSON) Client->>Controller: POST /api/trees/predict {"tableName": "provaC", "path": [1]} Controller->>Service: predictStep("provaC", [1]) Service->>Service: activeTrees.get("provaC") Service->>Tree: tree.navigate([1]) Tree-->>Service: PredictionStepDto (isLeaf=true, predictedValue=42.5) Service-->>Controller: PredictionStepDto Controller-->>Client: 200 OK (PredictionStepDto JSON)

9. Diagramma di Sequenza 2: Interazione Web UI Vaadin Flow (Porta 8081)

Diagram Error: No diagram type detected matching given configuration for text: sequenceDiagram autonumber actor Utente as Utente Web participant UI as MainView (Vaadin 8081) participant RESTClient as TreeRestClientService participant Backend as REST Controller (8080) Utente->>UI: Inserisce "provaC" in tableNameField e clicca learnBtn UI->>RESTClient: learnTree("provaC") RESTClient->>Backend: POST /api/trees/learn {"tableName": "provaC"} Backend-->>RESTClient: 200 OK (TreeResponseDto) RESTClient-->>UI: TreeResponseDto UI->>UI: rulesArea.setValue(treeRules) UI->>UI: renderStep(rootStep) -> popola pulsanti opzioni split UI->>Utente: Mostra regole e rami del nodo radice Utente->>UI: Clicca pulsante ramo (es. indice 1) UI->>UI: currentPath.add(1) UI->>RESTClient: predictStep("provaC", [1]) RESTClient->>Backend: POST /api/trees/predict {"tableName": "provaC", "path": [1]} Backend-->>RESTClient: 200 OK (PredictionStepDto) RESTClient-->>UI: PredictionStepDto UI->>UI: renderStep(step) -> rileva isLeaf == true UI->>Utente: Visualizza badge risultato predetto e riepilogo

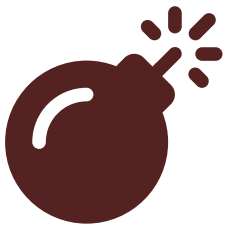
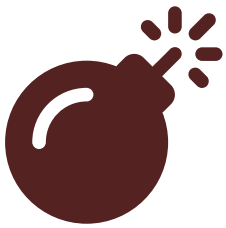
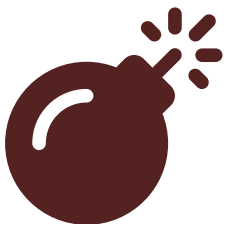
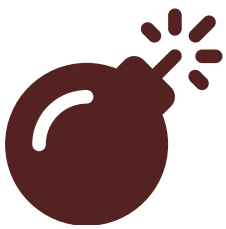
10. Diagramma di Sequenza 3: Protocollo Socket TCP Legacy (CLI - Porta 8080)

Diagram Error: No diagram type detected matching given configuration for text: sequenceDiagram autonumber actor Utente as Utente Console participant CLI as Main (Client CLI) participant KB as Keyboard participant Server as ServerOneClient (8080) participant DB as MySQL MapDB participant Tree as RegressionTree participant Disk as File System (.dmp) Utente->>KB: readInt() -> Scelta 1 (Apprendi da DB) Utente->>KB: readString() -> tableName "provaC" CLI->>Server: out.writeObject(0) [comando acquisizione] CLI->>Server: out.writeObject("provaC") Server->>DB: Query dati tabella 'provaC' DB-->>Server: Righe del training set Server-->>CLI: in.readObject() -> "OK" CLI->>Server: out.writeObject(1) [comando apprendimento] Server->>Tree: new RegressionTree(data) Server->>Disk: tree.salva("provaC.dmp") Server-->>CLI: in.readObject() -> "OK" Utente->>KB: readInt() -> Scelta 3 (Predizione) CLI->>Server: out.writeObject(3) [comando predizione] Server->>Tree: tree.predictClass(out, in) loop Fino a LeafNode Tree-->>CLI: in.readObject() -> stringa condizioni split CLI->>Utente: Visualizza query e opzioni del ramo Utente->>KB: readInt() -> inserisce indice ramo CLI->>Tree: out.writeObject(ramoScelto) end Tree-->>Server: return Double predictedValue Server-->>CLI: out.writeObject("OK") Server-->>CLI: out.writeObject(predictedValue) CLI->>Utente: Visualizza valore predetto a console

11. Diagramma di Deployment / Componenti (Porte 8080, 8081, 3306 MySQL MapDB)

Diagram Error: No diagram type detected matching given configuration for text: flowchart TB subgraph ClientHost ["Client Tier (Host Utente / Frontend)"] Browser["Web Browser (UI Vaadin Flow) http://localhost:8081"] ClientApp["Spring Boot Web Client (Porta 8081)
ClientApplication, MainView, TreeRestClientService"] CLIClient["Java CLI Client (Console)
Main, Keyboard"] end subgraph ServerHost ["Logic Tier (Server Host / Backend)"] direction TB Tomcat["Spring Boot Embedded Tomcat (Porta 8080)
ServerApplication, TreeController, TreeService, GlobalExceptionHandler"] SocketServer["MultiServer Socket TCP (Porta 8080)
MultiServer, ServerOneClient"] TreeEngine["Regression Tree Engine
RegressionTree, Data, DbAccess, TableData, TableSchema"] Tomcat --> TreeEngine SocketServer --> TreeEngine end subgraph DataHost ["Data Tier (Storage Host / Database)"] MySQL["MySQL Database Server 8.x (Porta 3306)
Database: MapDB | Utente: MapUser"] Disk["File System Locale
Archivi serializzati (*.dmp)"] end Browser -->|HTTP / WebSocket :8081| ClientApp ClientApp -->|HTTP REST POST / JSON :8080| Tomcat CLIClient -->|TCP Socket ObjectStreams :8080| SocketServer TreeEngine -->|JDBC Driver com.mysql.cj.jdbc.Driver :3306| MySQL TreeEngine -->|File I/O ObjectStreams| Disk SocketServer -->|File I/O ObjectStreams| Disk


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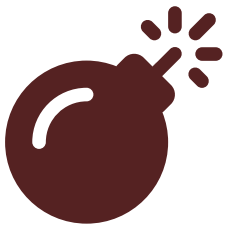
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